

Jerry (Ruiyang) Ji

707-849-2690 | rj378@cornell.edu | linkedin.com/in/ruiyangji | github.com/ruiyangji

Education

Cornell University — Ithaca, NY

Aug 2023 – Dec 2026

Bachelor of Science in Computer Science | **GPA: 3.8**

Relevant Coursework: Distributed Systems, Operating Systems, Machine Learning, Deep Learning, Parallel Computing, Database Management, Computer Networks, Data Structures

Experience

Meta — Software Engineering Intern | *Capacity Efficiency*

May 2026 – Aug 2026 • Menlo Park, CA

- Shipped **150+ production diffs** across 3 core milestones, independently scoping and architecting scalable backend features while leveraging AI developer tools to accelerate delivery cycles.
- Increased review adoption from **10% to 70%** across 3 orgs by engineering an automated **Chronos evaluation pipeline** to backtest 2,000+ historical capacity optimization proposals.
- Boosted evaluation accuracy by **20%** by adapting SkillOpt from Microsoft research and implementing automated reflection agent execution workflows for multi-step reasoning.
- Scaled review workflows on the Efficiency Hero platform for **3,000+ engineers**, enabling automated validation and auditing of **\$300M+** in quarterly compute savings.

PayPal — Software Engineering Intern | *Backend*

May 2025 – Aug 2025 • San Jose, CA

- Engineered **5+ production-ready backend features** for core billing and subscription services by collaborating cross-functionally across Agile sprints with QA, product, and frontend teams.
- Extended 10+ legacy API endpoints to support **20+ downstream microservices**, ensuring strict idempotency, contract compliance, and zero backward-compatibility regressions.
- Diagnosed and resolved distributed latency and failure bottlenecks across **8 services** using end-to-end distributed tracing, request context propagation, and logging telemetry.
- Elevated SonarQube code quality score from **75 to 90** by expanding unit and integration test suites and refactoring 20+ core service classes with standard Java design patterns.

Cornell Course Management System X — *Full Stack Software Engineer*

Sep 2023 – Apr 2024 • Ithaca, NY

- Maintained a **20+ year legacy codebase (100k+ LoC in Java)** for **8,000+ active students** by collaborating in Agile sprints, conducting rigorous code reviews, and resolving technical debt.
- Optimized server-side memory usage and reduced backend response times by **15%** by refactoring high-latency database queries and entity relationships utilizing Java WildFly and Jakarta JPA.

Projects

RL Cluster Scheduler | *Python, PyTorch, Gymnasium, PPO*

Jan 2026 – Present

- Built a discrete-event simulator to train RL agents for job scheduling; utilized **Action Masking** to enforce CPU/Memory constraints, increasing training convergence by **40%**.
- Engineered a **Permutation-Invariant Set-Attention Policy**, enabling **zero-shot generalization** to clusters 2x larger than training environments without performance degradation.

MealsApp: AI Meal Prep & Recipe Engine | *TypeScript, Gemini 2.5, Google APIs, PWA*

Sep 2026

- Built a serverless PWA powered by **Gemini 2.5 Flash** that dynamically adapts recipes and culinary substitutions against live pantry inventory with a calculated Flavor Match Score.
- Integrated multimodal receipt OCR and barcode lookup via Open Food Facts API (3.2M+ items), auto-generating batch cooking schedules into Google Calendar.

High-Performance SpMV Benchmark | *C, OpenMP, AVX2 SIMD*

Mar 2026 – May 2026

- Optimized SpMV kernels in C for CSR/ELLPACK/SELL-C-sigma formats using AVX2 SIMD and FMA, achieving **85%+ of theoretical peak memory bandwidth**.
- Conducted Roofline Analysis to identify memory-bound bottlenecks and implemented row-sorting (sigma-permutation) to balance SIMD efficiency against irregular graph padding.

Technical Skills

Languages: Java, Python, C, C++, Go, SQL, TypeScript, Scala, Hack, PHP, Bash

Systems & Frameworks:	Kafka, Spark, Spring Boot, Apache Iceberg, Redis, JunoDB, OpenMP, Microservices
Cloud & DevOps:	AWS, Kubernetes, Docker, Jenkins, CI/CD, Terraform, Linux, Maven, Gradle, OpenTelemetry
Data & Storage:	PostgreSQL, MySQL, MongoDB, DynamoDB, Jakarta JPA, Hibernate, Connection Pooling